ETHEREUM CRYPTOGRAPHY

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HMWK 2 – FROM WALLET SEED TO ETHEREUM ADDRESS

1. Pipeline

Mnemonic → (BIP-39) → 512-bit seed → (BIP-32/BIP-44)→ private key → (secp256k1)→ public key→ (Keccak-256)→ hash→ last 20 bytes = address→ (EIP-55)→ checksummed address.

Standard	Role
BIP-39	Generates entropy (128–256 bits) and appends a checksum (1 bit per 32 bits of entropy). The result is split into 11-bit groups, each an index into a list of 2048 words: 12–24 words. The seed is PBKDF2HMAC-SHA512 (2048 iterations, salt "mnemonic" + optional passphrase), 512 bits.
BIP-32	Hierarchical deterministic wallets. HMAC-SHA512 with key "Bitcoin seed" applied to the seed gives a master key (32 bytes) and a chain code (32 bytes). Children (normal or hardened) are derived from a parent: one seed yields unlimited keys.
BIP-44	Standard path structure: m / purpose' / coin_type' / account' / change / index. For Ethereum: m/44'/60'/0'/0/0 (60 = ETH). Ensures interoperability between wallets.
Secp256k1	Curve $y^2 = x^3 + 7$ over a 256-bit prime. Q = kG is easy, k from Q is infeasible (ECDLP). Also used for signatures (ECDSA).
Keccak-256	Hash: the address is the last 20 bytes of the Keccak-256 of the public key (64 bytes, without the 0x04 prefix). It shortens the key and hides it until the first signature.
EIP-55	Checksum encoded in letter case: the Keccak-256 of the lowercase address (hex text) is computed; each alphabetic digit is uppercase if the corresponding hash nibble is ≥ 8.

2. Concepts to distinguish

- **Key generation vs key derivation**: generation= creating random entropy (the mnemonic); derivation= deterministic computation of keys from an existing secret (BIP-32/44). Same input, same output.
- **Public key cryptography:** a pair (k,Q) with a one way relation; used to sign and verify (secp256k1).

- **Hashing**: keyless function, fixed-length output, preimage and collision resistant (Keccak-256; SHA-512 inside HMAC and PBKDF2).
- **Checksum:** redundancy to detect accidental errors; it does not protect against an attacker (EIP-55 and the checksum bits of the mnemonic).
- **Encoding:** reversible representation with no secret (hexadecimal, word list, upper/lower case). It provides no security.

3. Historical context

Year	Event
1977	RSA, public-key cryptography (elliptic-curve cryptography follows in 1985)
2001	SHA-2 / SHA-512 standardized by NIST
2008-2012	Keccak is proposed and wins the NIST SHA-3 competition (2012)
2012	BIP-32, HD wallets
2013	BIP-39, mnemonic code
2014	BIP-44, multi-account paths
2015	SHA-3 standardized (FIPS 202); Ethereum launches with "pre-standard" Keccak
2016	EIP-55, mixed-case checksum

4. SHA-2/ SHA-512 vs Keccak/SHA-3

Both are cryptographic hash functions, but with different designs: SHA-2 uses the Markle-Damgård construction (iterated compression), Keccak uses the sponge construction on a 1600-bit permutation. The independent desing makes SHA-3 a fallback should SHA-2 be weakened.

Ethereum uses KECCAK-256, which is not identical to SHA-256: during standardization NIST changed the padding/domain – separation byte (Keccak: 0x01; SHA-256: 0x06).

5. Why several techniques instead of one algorithm

- **BIP-39/32/44 →** key management and derivation: human-readable backup, many accounts from one seed, interoperability.
- **secp256k1 →** public/private key relationship and digital signatures: only public-key cryptography allows verification without revealing k.
- **Keccak-256 →** hashing: a short (160-bit), uniformly distributed address, with an extra layer between Q and the public.

- **EIP-55 →** checksum: detects typos without changing the address format or length.

One algorithm is not enough: a hash has no public keys or signatures; an elliptic curve neither compresses nor diffuses bits; a checksum offers no security. This is the principle of separation of concerns: each component is simple, standardized and replaceable.

6. Statistical properties

- **Mnemonic**: uniform entropy produced by a secure random generator; 12 words = 128 bits (+4 checksum), 24 words = 256 bits (+8).
- **PBKDF2/HMAC-SHA512 and Keccak**: outputs indistinguishable from random, avalanche effect (1 input bit flips about 50% of output bits), collision resistance (160bit address: birthday collision at about 2^{80}).
- **Public key kG**: no avalanche effect; it has algebraic structure ((a + b)G = aG + bG). What matters is the hardness of the ECDLP.
- **EIP-55**: with about 15 letters on average, roughly one error in 2^{15} (about 30,000) goes undetected.

7. Experiment with a wallet

By creating a test wallet (MetaMask, Rabby, Rainbow or similar; with no funds and never with a real mnemonic) one can observe: the word list (BIP-39); that adding new accounts changes the last index of the path (BIP-44); the address shown with upper and lower case letters (EIP-55); and that importing the same mnemonic into another wallet reproduces the same addresses, confirming the determinism of BIP-32/44.